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An AI-Ready Air-to-Ground Channel Model for HAPS–UAV Access Networks

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ABSTRACT This paper develops an AI-ready air-to-ground channel model for a HAPS–UAV–Ground access network and uses it to train a learned user–platform association policy. We implement (i) a free-space and ITU-R atmospheric (rain, fog, gas, scintillation) channel model for the HAPS backhaul/feeder link (38 GHz) and the HAPS/UAV service link (2 GHz), (ii) a building-blockage line-of-sight model for the UAV access hop, and (iii) the resulting downlink SINR under a 3-HAPS equilateral-triangle constellation, including multi-HAPS co-channel interference. This channel model reveals spatial SINR heterogeneity: at a fixed 50 km radius from the serving HAPS, SINR varies by up to 9.94 dB with receiver angle alone – more than the 2–3 dB cost of doubling the link distance – because of co-channel interference from the other two platforms, showing that a distance-only link budget is insufficient and motivating a learned association policy. Three rule-based policies (random, max-SINR, max-throughput) are evaluated first as baselines, confirming that association choice alone sharply affects outage and fairness. An LSTM forecasts 38 GHz feeder rain attenuation up to six hours ahead from historical weather; on held-out data it matches or beats persistence on RMSE at four-to-six-hour horizons and on MAE at five-to-six-hour horizons, and averaging the LSTM and persistence forecasts further improves RMSE over either alone. A Deep Q-Network (DQN) agent that consumes this forecast, together with per-platform SINR, distance and battery state, learns a six-way (direct or UAV-relay, per HAPS) user–platform association policy. Across five training experiments and 50 held-out days, the DQN earns higher reward than a perfect-information max-throughput rule and a max-SINR rule, with zero outage and fairness statistically indistinguishable from max-throughput.

INDEX TERMS HAPS, UAV, non-terrestrial networks, air-to-ground channel, LoS probability, SINR, multi-node interference, UAV relay, backhaul link, rain attenuation, LSTM forecasting, deep reinforcement learning

I. INTRODUCTION

A High-Altitude Platform Station (HAPS) is a stratospheric, solar-powered platform at approximately 20 km altitude with a fixed service footprint of up to 100 km radius [1], [2]. It connects to the ground over three link types:

- **Service link** (HAPS to UE, 2–2.2 GHz [30]): direct, long-range connectivity without specialized ground hardware.
- **Backhaul link** (HAPS to Gateway, 38.0–39.5 GHz Ka-band, WRC-19 [29]): high-capacity ground-gateway connectivity.
- **Relay link** (HAPS to UAV to UE): a two-hop backup path for users with a poor direct link – a 38 GHz feeder hop (HAPS–UAV, same band as the backhaul) followed by a 2 GHz access hop (UAV–UE) from a UAV hovering at lower altitude.

The 38 GHz backhaul/feeder links incur significant atmospheric loss (rain, fog, gas), while sitting in a spectral valley between the 22.235 GHz water-vapor and 60 GHz oxygen absorption resonances [6]. The 2 GHz service/relay links are resilient to atmospheric effects [4] but sensitive to building blockage. Because the 100 km service radius pushes many links to low elevation angles,

slant-path lengths and the resulting atmospheric loss grow accordingly [4].

Distance-only link budgeting, however, misses a second, larger effect: with three HAPS sharing spectrum, co-channel interference from the other two platforms makes SINR depend strongly on *angular position*, not just distance. A static analysis cannot react to this, or to time-varying rain fades on the 38 GHz backhaul. This motivates an AI-ready model: a validated channel model, an LSTM rain forecaster, and a DQN agent that learns real-time user–platform association.

Research flow. The remainder of this paper follows three stages: (1) a *channel and attenuation model* – free-space path loss, ITU-R rain/fog/gas/scintillation, and multi-HAPS interference (§III–§V), characterised numerically in §VI; (2) *baseline association policies* – random, max-SINR, and max-throughput rules, evaluated on that channel model to establish how much association choice alone matters (§VI-E); and (3) a *learned association policy* – an LSTM rain forecaster (§VII) feeding a DQN agent (§VIII), evaluated against the stage-2 baselines (§IX).

The results follow the same three stages: (1) a reproducible ITU-R-based channel model for the HAPS, UAV and gateway links, including a quantitative characterisation of multi-HAPS interference showing where it limits the 2 GHz service link and why the 38 GHz backhaul does not, and the HAPS→UAV→UE relay path as an alternative to the direct service link; (2) three rule-based association baselines (random, max-SINR, max-throughput) evaluated on that model; and (3) an LSTM rain forecaster benchmarked out-of-sample against persistence, feeding a DQN association agent trained on 2024 weather and evaluated on held-out 2025 days against the stage-2 baselines.

II. RELATED WORK

Building-blockage/LoS modeling (2 GHz access tier). Holis and Pechac [18] give the elevation-dependent building-blockage shadowing model this paper adopts for the 2 GHz HAPS/UAV access tier, originally formulated for a single serving platform with no co-channel interference term; ITU-R P.1410 [11] offers a simpler elevation-angle-only LoS baseline that the Holis–Pechac model refines with building geometry. The 38 GHz backhaul/feeder link does not need this model: its near-nadir geometry (§III) avoids building obstruction, so atmospheric loss (rain/fog/gas, §IV) is the dominant impairment there instead.

Multi-HAPS co-channel interference. Arani *et al.* [17] study HAPS–UAV heterogeneous networks with a DRL controller, closest in spirit to the association problem here, but evaluate a single HAPS–UAV pair with no co-channel interference term. Liu *et al.* [33] optimize sum-rate and worst-case SINR across multiple cooperating HAPS and ground base stations via joint user association and beamforming, but treat the channel as static and do not model time-varying rain fades or an onboard forecaster; our channel model instead quantifies the *angular* SINR heterogeneity this cooperative setting would need to counteract, and complements it with the learned, weather-aware policy their static optimization does not provide.

Contrast with Arani *et al.* [17]. Because this is the closest prior work, we make the delta explicit. (i) *Interference:* their DRL controller is evaluated on a single HAPS–UAV pair, so it has no co-channel term to learn around; our 3-HAPS geometry shows angular position alone swings SINR by 9.94 dB at fixed range (§V), a degree of freedom their setting cannot expose. (ii) *Weather coupling:* their feeder/access channel is time-invariant, so the controller cannot anticipate fades; our DQN consumes a six-horizon LSTM rain forecast (§VII) specifically because the feeder link is not static. (iii) *Reward structure:* we jointly penalize outage, fairness, and per-platform battery depletion (§VIII, Eq. (9)) in a single reward, rather than optimizing throughput alone, so the comparison against rule-based max-throughput/max-SINR baselines (§IX) isolates what the extra state (forecast, battery, fairness) buys over a narrower throughput-only objective.

Learned resource allocation and association in NTN. Naous *et al.* [32] survey reinforcement learning across non-terrestrial platforms (satellites, HAPS, UAVs) and note that most RL-based association work either assumes a static channel or omits weather-driven feeder-link variability – the gap this paper's LSTM-fed DQN targets. Foundational DQN [25] and RL theory [26] underpin the agent design of §VIII; PPO [27] was considered but DQN suits this paper's small discrete (six-action) space better, at lower sample complexity. Fairness is quantified with the Jain index [28], and per-link SINR is combined in the linear domain following standard practice [22] rather than by averaging in dB.

Rain attenuation forecasting. Cornejo *et al.* [34] train an LSTM on synthetic, ITU-R-derived rain-attenuation traces for satellite links, reporting improved short-term prediction accuracy over classical models; this paper instead trains on several years of real, multi-source weather data (NASA POWER, METAR, ERA5, 2023–2026) for a specific HAPS feeder-link geometry and benchmarks against a persistence baseline, which (§VII) proves non-trivial to beat at short horizons. The resulting rain margins are consistent with the Ka-band FSS rain-margin guidance in ECC Report 156 [20] (§IV-A).

Survey context. The recent survey by Svistunov *et al.* [31] catalogs HAPS channel modeling, interference control, and AI-driven resource allocation at a high level, and the broader survey by Bakambekova *et al.* [19] on AI for space-air-ground integrated networks identifies weather-aware, cross-layer learning as an open direction at the SAGIN level; this paper provides one of the quantitative instances – a validated channel model and a

TABLE 1: Simulated network configuration (AGL = above ground level)

Node	Altitude	Count	Spacing
HAPS	20 km	3	Equilateral 65 km, 1 server, 2 interferers
UAV relay	500 m AGL	1	Static (no mobility)
Gateway	50 m AGL	1	Centroid of HAPS triangle
Users	1.5 m (hand-held)	N (configurable)	Uniform random in coverage

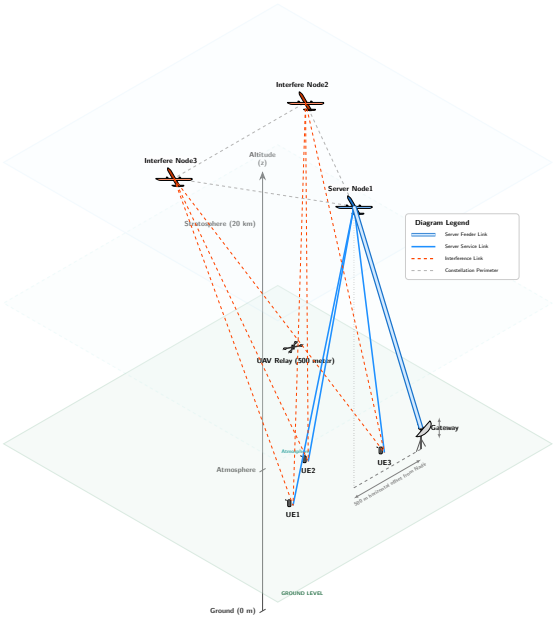


FIGURE 1: Network topology: three HAPS nodes and one UAV relay serve ground users over an access layer; a single ground gateway aggregates the backhaul links into the core network.

trained, evaluated association policy – that both surveys point to as needed.

Positioning. None of the above combines all three elements this paper does: (i) an ITU-R-validated, multi-HAPS interference-aware channel model showing that angular position can dominate distance in SINR, (ii) an out-of-sample rain forecaster feeding (iii) a DQN association policy evaluated against rule-based baselines on real held-out weather. This combination, rather than any single component in isolation, is the contribution.

III. SYSTEM TOPOLOGY AND CHANNEL MODEL

Three HAPS nodes and one UAV relay serve ground users (access layer); a single ground gateway aggregates all three backhaul links (backhaul layer) into the core network (Table 1, Fig. 1). Elevation angle and slant-path length follow directly from each link's altitude pair and horizontal offset: the backhaul link (HAPS to a near-nadir gateway) and the HAPS–UAV feeder hop are both near-zenith (≈88.5–89.9° at nadir); the UAV–UE access hop is dominated by horizontal offset and building blockage rather than altitude; and the direct HAPS–UE service link spans the full service radius, from near-zenith at nadir to low elevation at the 100 km edge.

Channel model. Three propagation models apply to this topology: (i) free-space path loss (FSPL), common to all links; (ii) ITU-R rain, fog, gas and scintillation losses, applicable to the 38 GHz backhaul and feeder links; and (iii) a building-blockage LoS/NLoS probability model (Holis & Pechac [18]), applicable to the 2 GHz UAV/HAPS access-tier links.

The ITU-R P.530 free-space path loss [3] is

$$L_{\mathrm{FSPL}} = 32.44 + 20\log_{10} f_{\mathrm{MHz}} + 20\log_{10} d_{\mathrm{km}}\ [\mathrm{dB}], \quad (1)$$

where f_{MHz} is the carrier frequency and d_{km} the link distance. This is algebraically the wavelength form $L_{\mathrm{FSPL}}(h,r) = 20\log_{10}(4\pi d_{h,r}/\lambda)$ used later in the multi-HAPS SINR formulation (§V): substituting $\lambda = c/f$ and converting units folds the resulting constant into the 32.44 dB term above.

IV. ATMOSPHERIC AND PROPAGATION MODELS

A. 38 GHZ BACKHAUL/FEEDER LINK

Rain attenuation γ_R (dB/km) follows the ITU-R P.838 power law [5]

$$\gamma_R = k(f,\theta)\, R^{\alpha(f,\theta)} \quad [\mathrm{dB/km}], \quad (2)$$

with $k \approx 0.392$, $\alpha \approx 0.869$ at 38 GHz for the near-nadir (≈ 89.86°) HAPS–gateway geometry (horizontal polarisation), giving $\gamma_R \approx 2.9$ dB/km at 10 mm/h and ≈ 6.4 dB/km at 25 mm/h. Multiplying by the ITU-R P.618 effective slant length L_{eff} [4] gives $A_{\mathrm{rain}} = \gamma_R L_{\mathrm{eff}}$. For Delhi's $R_{0.1} = 15.7$ mm/h rain rate (ITU-R P.837 [10]), this gives ≈ 19 dB of rain loss on the near-nadir gateway path. Against the clear-sky backhaul SINR of ≈ +38 dB (§V) and ≈ 0.4 dB of additional gas loss, the resulting link margin is $38 - 19 - 0.4 \approx 18.6$ dB at 99.9 % availability – consistent with the 15–25 dB Ka-band rain margins reported for FSS links by ECC Report 156 [20] – so the 38 GHz backhaul therefore closes comfortably at the design availability target.

Fog/cloud loss (ITU-R P.840 [7]) and gas absorption (ITU-R P.676 [6]) are modeled as specific attenuations times slant length, $A_{\mathrm{cloud}} = \gamma_c(f,M,T)L_{\mathrm{slant}}$ and $A_{\mathrm{gas}} = (\gamma_o + \gamma_w)L_{\mathrm{slant}}$; both are small at 38 GHz (fog typically < 0.2 dB; clear-sky gas ≈ 0.2–0.3 dB/km, versus ∼15–20 dB/km at the 60 GHz oxygen resonance). Tropospheric scintillation (ITU-R P.618 Annex 1 [4]) is

$$A_{\mathrm{scint}}(p) = a(p)\cdot\sigma_{\mathrm{pre}}(f,\theta,\mathrm{climate}) \quad [\mathrm{dB}], \quad (3)$$

growing with frequency and shrinking with elevation. Total backhaul/feeder slant-path attenuation sums all terms:

$$L_{\mathrm{total}} = L_{\mathrm{FSPL}} + A_{\mathrm{rain}} + A_{\mathrm{cloud/fog}} + A_{\mathrm{gas}} + A_{\mathrm{scint}}. \quad (4)$$

B. 2 GHZ SERVICE/RELAY LINK

Unlike the 38 GHz links, the 2 GHz band is resilient to atmospheric effects but sensitive to building blockage. For a UAV or HAPS node u at horizontal distance $r_{u,k}$ from user k, the LoS probability follows the building-blockage product series of Arani et al. [17], parameterised by Holis & Pechac [18]:

$$P_{u,k}^{\mathrm{LoS}} = \prod_{n=0}^{J} \left[1 - \exp\left(-\frac{1}{2\xi^2} \left(h_u - \tfrac{n+\frac{1}{2}}{J+1}(h_u - h_k) \right)^2 \right) \right], \tag{5}$$

with $J = \lfloor \frac{r_{u,k}}{1000}\sqrt{\alpha\beta} - 1 \rfloor$ building layers, and environment-dependent parameters (α, β, ξ) (e.g. suburban $0.1, 750, 8$ m; urban $0.3, 500, 15$ m; dense urban $0.5, 300, 20$ m; high-rise $0.5, 300, 50$ m) [18]. This single-relay topology (Table 1) needs only a 3-HAPS interference sum; no UAV-to-UAV term is required.

Each propagation state $z \in \{\mathrm{LoS}, \mathrm{NLoS}\}$ adds a log-normal shadowing term, averaged in linear gain (not dB, since the two states can differ by up to ∼17 dB in dB-domain averaging):

$$\overline{g} = 10^{-\mathrm{FSPL}/10} \left[P_{\mathrm{LoS}}\, e^{\frac{(a\sigma_{\mathrm{LoS}})^2}{2}} + P_{\mathrm{NLoS}}\, e^{-a\mu(\theta) + \frac{(a\sigma(\theta))^2}{2}} \right],$$
$$\overline{\mathrm{PL}} = -10\log_{10}\overline{g}, \qquad a = \tfrac{\ln 10}{10}. \tag{6}$$

The downlink SINR at user k from node u is then

$$\gamma_{u,k} = \frac{p_u\, g_{u,k}}{\sum_{u' \neq u} p_{u'} g_{u',k} + \sigma_0^2}, \qquad g_{u,k} = 10^{-\overline{\mathrm{PL}}/10}. \tag{7}$$

V. MULTI-HAPS INTERFERENCE, SINR, AND THE RELAY PATH

With three HAPS sharing spectrum, a serving HAPS h provides the desired signal while the other two contribute co-channel interference at the same receiver r (gateway, UAV, or UE):

$$\gamma_{h,r} = \frac{P_h G_h G_r g_{h,r}}{N_r + \sum_{h' \neq h} P_{h'} G_{h'} G_r g_{h',r}}, \tag{8}$$

with channel gain $g_{h,r} = 10^{-L_{h,r}^{\mathrm{tot}}/10}$ from the total propagation loss of §IV. This same formulation instantiates the backhaul link (r = gateway), the feeder hop (r = UAV), and the direct service link (r = UE); the UAV–UE access hop uses UAV transmitters/receivers in place of HAPS and is interference-free with a single relay.

Backhaul link. The gateway sits near the serving HAPS's nadir, while the two interfering HAPS are ≈65 km away and seen off-boresight through sidelobes (ITU-R F.699, floored at −10 dBi). With directional antenna gains included ($G_h = 13.8$ dBi, $G_g = 39.7$ dBi [14]), the interference term is ≈ −35 dB below the noise floor, so the backhaul SINR is ≈ +38 dB and **noise-limited**, essentially flat across the 10–500 m gateway-offset range considered.

Feeder hop (HAPS→UAV). The same near-nadir geometry applies to the UAV, so this hop is also noise-limited: +37.6 dB at nadir falling to +22.6 dB at 100 km HAPS–UAV distance as elevation drops to 11°, using $G_h = 13.8$ dBi and a UAV-mountable mmWave patch array $G_u = 30.7$ dBi [15].

Service link (HAPS→UE). Here the picture is different. A receiver at horizontal distance r from the serving HAPS's nadir and the two interferers at ≈65 km produces an **angularly-dependent** SINR: at a fixed $r = 50$ km, SINR ranges from −7.40 dB toward an interferer ($\theta = 0°$) to +2.54 dB away from the constellation ($\theta = 210°$) – a 9.94 dB spread – mapping to a good sector (≈120°–290°) and a bad sector (≈330°–75°) (Fig. 6, §VI). Averaged over all azimuths, mean service-link SINR falls from +6.3 dB at 10 km to −4.3 dB at 100 km, while the angular min–max band widens from 1.95 dB to a peak of 11.94 dB at 75 km (Fig. 7, §VI) – beyond ∼30 km, angular position dominates over distance decay. This spread (9.94 dB at fixed distance) exceeds the ∼2–3 dB cost of simply doubling the link distance, so a distance-only link budget cannot characterise the interference environment, and the network needs TDMA/FDMA/beamforming/power control, or a learned association policy, to manage it.

A. RELAY PATH ANALYSIS (HAPS→UAV→UE)

The relay path's end-to-end capacity is the decode-and-forward bottleneck of its two hops, $C_{\mathrm{relay}} = \min(C_{\mathrm{haps-uav}}, C_{\mathrm{uav-ue}})$. Because the UAV hovers close to the user it serves, the access hop stays near +33 dB SINR regardless of HAPS distance, while the feeder hop (above) stays well above a +8 dB design floor (16-QAM/CQI-7 minimum-MCS SINR at 10 % BLER plus a 3 dB margin [13]) at every distance tested. The direct service link, by contrast, is interference-limited and falls from +7.3 dB to −9.1 dB over the same range (§V). The relay therefore **outperforms the direct link at every distance**: end-to-end capacity is 11.1 bits/s/Hz at nadir (access-hop-limited) and 7.5 bits/s/Hz at 100 km (feeder-hop-limited), against the direct link's 2.7 and 0.2 bits/s/Hz. This motivates including the UAV relay as an association option for the learned policy of §VIII.

VI. SIMULATION RESULTS

All results use a Python simulation stack (`NumPy/pandas/Matplot`
atmospheric and scintillation losses use the ITU-Rpy (`itur`) [24] library's implementation of the relevant ITU-R recommendations.

A. WHY 38 GHZ: FREQUENCY SELECTION

This subsection validates the 38 GHz backhaul/feeder choice already fixed in §III (WRC-19 allocation). It is included to address a natural reader question: why not simply pick a lower-loss frequency instead of designing around rain, fog, and gas attenuation at 38 GHz?

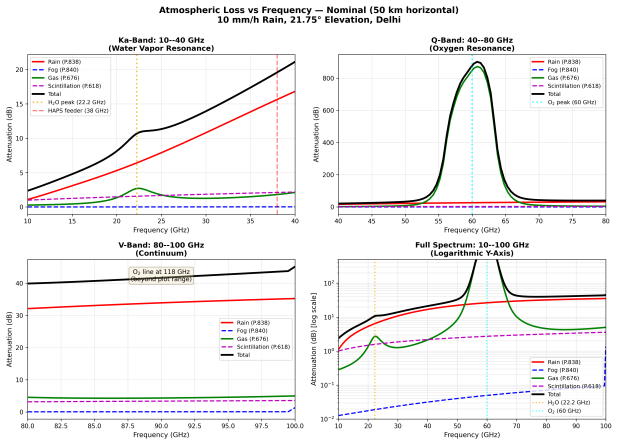


FIGURE 2: Atmospheric loss (rain, fog, gas) vs. frequency across the Ka/Q/V bands for the Delhi gateway path at 10 mm/h rain.

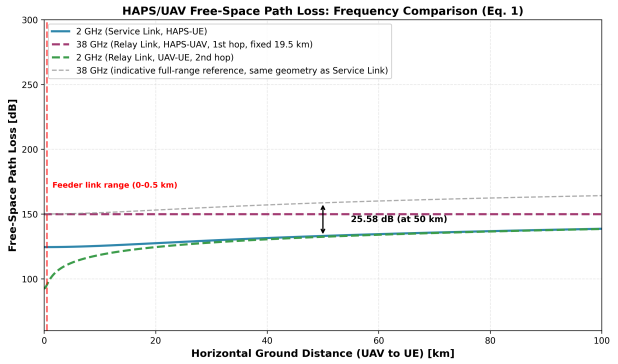


FIGURE 3: FSPL vs. distance (Eq. (1)) – same functional form for 2 and 38 GHz; the 38 GHz feeder sits ≈25.6 dB higher, purely from the $20\log_{10}(38/2)$ frequency term.

Rain loss (ITU-R P.838/P.618 [4], [5]) is the dominant term and grows roughly linearly with frequency, up to ≈19 dB at 99.9 % availability. Fog/cloud (ITU-R P.840 [7]) is negligible below 10 GHz. Gas absorption (ITU-R P.676 [6]) has two resonance peaks to avoid: water vapor at 22.2 GHz and oxygen at 60 GHz, where total loss exceeds 800 dB over 50 km and the band is unusable. **38 GHz sits in the valley between these peaks** (≈0.2–0.3 dB/km clear-sky gas loss) – this, not just available bandwidth, is why it was chosen over a frequency closer to either resonance.

B. PATH LOSS AND SINR VS. DISTANCE

C. ANGULAR INTERFERENCE SECTORS AND SCINTILLATION

The good/bad-sector result of §V (Fig. 6) is one half of the angle- and weather-dependent picture; scintillation (Eq. (3)) is the other. At 30° elevation, $p = 0.01\,\%$, scintillation fade is ≈0.49 dB at 38 GHz versus ≈0.09 dB at 2.1 GHz – growing with frequency and shrinking with elevation. Both effects favor positioning the UAV relay

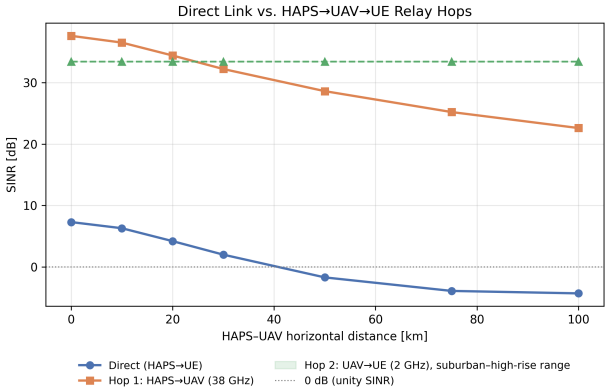


FIGURE 4: SINR vs. HAPS–UAV horizontal distance: the direct service link falls below 0 dB past ≈40 km, while both UAV-relay hops (feeder 38 GHz, access 2 GHz) stay well above it at every distance tested (§V-A).

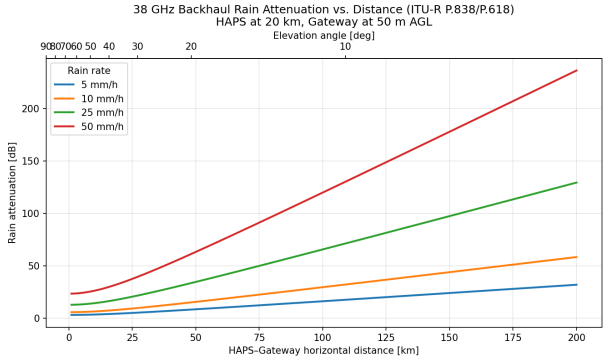


FIGURE 5: 38 GHz backhaul rain attenuation vs. HAPS–Gateway horizontal distance (ITU-R P.838/P.618), grows faster than linear past ≈100 km as elevation drops and the slant path lengthens (§IV-A).

near nadir (higher elevation) when feasible, especially on the HAPS–UAV feeder hop, which also tends to fall in the interferer-facing sector's complement. This is not, however, a single clean optimization: each HAPS can simultaneously be a serving node (near-nadir, low-loss, for its own users) and an interferer for the other two HAPS' users – jointly optimizing positioning across both roles is left to future work (§X).

D. WHY LEARNING IS NEEDED: SPATIAL SINR HETEROGENEITY

At a fixed 50 km radius, SINR varies by up to 9.94 dB with angular position (§V), while doubling the link distance costs only ∼2–3 dB – angular position affects SINR more than distance does. A simple "always pick max-SINR" rule cannot anticipate these angular patterns or adapt as weather and battery state change, motivating a learned, real-time association policy.

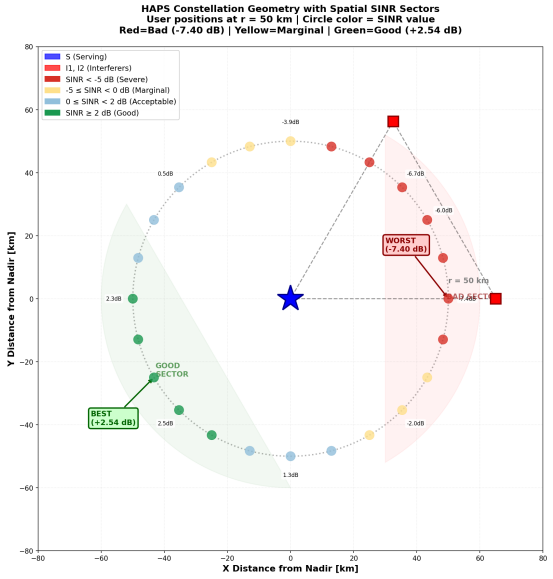


FIGURE 6: HAPS constellation geometry (65 km equilateral triangle: S at origin, I1 at 0°, I2 at 60°) with spatial SINR sectors at $r = 50$ km. Good sector (green, ∼120–290°) vs. bad sector (red, ∼330–75°). Best +2.54 dB at 210°; worst −7.40 dB at 0°.

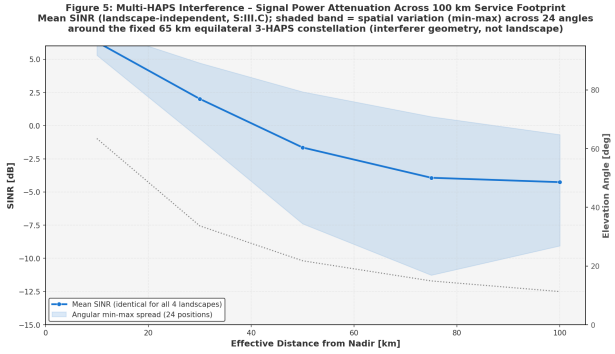


FIGURE 7: Mean SINR and angular min–max spread across the 100 km service footprint: falls from +6.3 dB at 10 km to −4.3 dB at 100 km, while the spread widens to 11.94 dB at 75 km.

E. STAGE 2: BASELINE ASSOCIATION POLICIES

Before introducing a learned policy, three rule-based (non-learned) association policies are evaluated on the channel model above, to establish how much association choice alone – without any learning – already changes outcomes. Each of N users chooses one of six options every hour: a direct link to HAPS1–3, or a UAV relay via HAPS1–3 (§V-A).

- **Random**: uniform-random choice among the six options each hour (sanity floor).
- **Max-SINR**: always the best-SINR *direct* link (Eq. (7)); cannot use the relay, no memory of battery or fairness state.

TABLE 2: LSTM against persistence on 2026 (6557 forecast hours, no 2026 data in training): MAE/RMSE in dB per horizon. Bold marks the better method.

Horizon	LSTM MAE	Persist. MAE	LSTM RMSE	Persist. RMSE
+1 h	0.20	**0.12**	0.65	**0.53**
+2 h	0.27	**0.20**	0.86	**0.75**
+3 h	0.31	**0.26**	0.96	**0.90**
+4 h	0.33	**0.30**	**1.00**	1.02
+5 h	0.34	0.34	**1.00**	1.11
+6 h	**0.35**	0.36	**1.02**	1.17

- **Max-throughput**: the highest-throughput option among all six every hour, assuming perfect, instantaneous knowledge of every option's SINR and capacity – a theoretical upper bound, not a deployable policy.

These three policies are carried forward as the baselines in §IX, where they are compared against the learned LSTM+DQN policy of §VIII on the same held-out days and the same six-action set.

VII. LSTM RAIN-ATTENUATION FORECASTER

The DQN must choose an association at hour t before that hour's rain is observed, so it needs a forecast of the feeder loss. An LSTM is trained to map the past 24 hours of weather (14 features: temperature, humidity, pressure, wind, precipitation, cloud amount, visibility, rain/fog codes, cloud liquid water, and sine/cosine encodings of time-of-day and season, from hourly NASA POWER, METAR and ERA5 data for Delhi, 2023–2026) to the next six hourly values of HAPS-to-gateway rain attenuation (38 GHz, 89.86° elevation, ITU-R P.838/P.618) – the only link in this model where rain is a significant, time-varying impairment (§IV-B); the 2 GHz service/relay link's rain, fog, and gas loss are negligible, so no forecast is needed there. Forecasts are always evaluated out-of-sample (e.g. 2026 forecasts from a model trained only on 2023–2025), and a forecast for hour t is issued at $t-1$, so the agent never sees the rain of the hour it is acting on. In deployment this would be a continuous, incremental feed, with the model updating as new weather data arrives.

Persistence is more accurate on RMSE at +1 to +3 h and on MAE at +1 to +4 h; the LSTM is more accurate on RMSE at +4 to +6 h, and on MAE it ties persistence at +5 h and wins at +6 h (Table 2). The DQN's state uses all six forecast horizons regardless, since the gain is not uniform across horizons but the longer horizons matter most for pre-emptive rerouting. **Observation:** simply averaging the LSTM and persistence forecasts beats both alone on RMSE (0.511 vs. 0.645 LSTM-only and 0.530 persistence-only) – forecast ensembling is a promising direction, noted further in §X.

VIII. DQN ASSOCIATION AGENT

A. PER-USER MDP FORMULATION

Each of the three platforms can serve a user directly or through the UAV relay, giving six actions $a_k \in \{0, \ldots, 5\}$

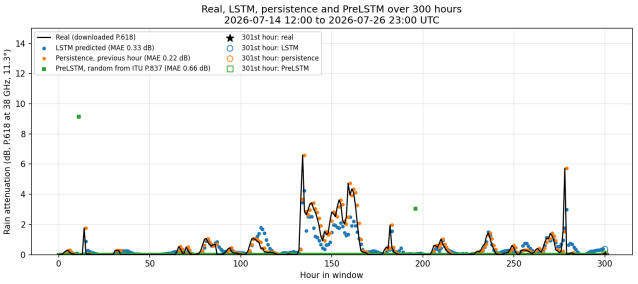


FIGURE 8: Real, LSTM, persistence and a synthetic PreLSTM baseline over a 300-hour window in July 2026 (the window with the most rain). LSTM: one-hour-ahead forecast. Persistence: previous hour's value. PreLSTM: random hourly rain rates drawn from ITU-R P.837 statistics, converted via P.618.

(actions 0–2: direct HAPS1–3; actions 3–5: UAV relay via HAPS1–3). With N users, a centralized MDP over joint assignments is intractable, so the problem decomposes into per-user MDPs: each user independently chooses a platform, and reward is computed after all assignments.

With a single, static relay UAV (Table 1), choosing *which* HAPS feeds the UAV's feeder hop is folded into this same per-user action set, effectively treating the UAV as if it were just another user requesting a serving HAPS. Only the *shape* of that choice is shared (pick one of three HAPS) – the channel model evaluating it is not: the UAV's feeder hop still uses the 38 GHz, ITU-R-P.618 feeder model of §IV-A (its own antenna gains and atmospheric terms), never the 2 GHz building-blockage model of §IV-B that a direct user's choice is evaluated against. Sharing the action's shape, not its physics, is a fair simplification when there is only one UAV to assign. This does not generalise to a UAV fleet: at fleet scale, UAV-to-HAPS assignment (an O(fleet size) problem, e.g. the ∼200-UAV deployment of §X) and UE-to-UAV/HAPS access (an $O(N)$ problem) become two separably-sized assignment problems that cannot stay bundled into one per-user action, since the UAV side would then be a shared resource contended for by many users' decisions at once rather than belonging to any single user's action space.

State (per user, 21-dimensional): SINR to each platform (dB, Eq. (7)), horizontal distance to each (km), each HAPS's battery state of charge (%, §VIII-B), the UAV-relay capacity to each platform (bits/s/Hz, §V-A), rainfall rate, visibility, time-of-day, and the six LSTM rain forecasts $\hat{A}_t, \ldots, \hat{A}_{t+5}$ (§VII), issued at $t-1$ so no rain of hour t enters the state. Tropospheric scintillation (§IV-A) is not included as a state feature – it is a smaller, time-varying effect ($\leq$0.6 dB at 38 GHz, $p = 1\,\%$) whose forecasting is left to future work.

Reward. For user k with linear SINR $\gamma_k = 10^{\mathrm{SINR}_{k,\mathrm{dB}}/10}$ on its chosen link,

$$r_k = \omega_1 T_k c_{\mathrm{rain}} + P_k^{\mathrm{sev}} + P_k^{\mathrm{deg}} + \omega_2 J_{\mathrm{fair}} - \omega_3 P_{\mathrm{eng}} + P_{\mathrm{LB}},$$
$$T_k = \log_2(1+\gamma_k), \qquad c_{\mathrm{rain}} = 10^{-A_{\mathrm{rain}}/10} \in [0.05, 1],$$
$$J_{\mathrm{fair}} = \frac{(\sum_k \gamma_k)^2}{N \sum_k \gamma_k^2}, \qquad P_{\mathrm{eng}} = \frac{1}{3}\sum_{p=1}^{3}\Big(1 - \frac{\mathrm{SoC}_p}{100}\Big), \tag{9}$$

with $\omega_1 = 0.5, \omega_2 = 0.3, \omega_3 = 0.2$; c_{rain} de-rates every user's throughput credit alike, whether served directly or via the UAV relay, because all traffic – direct or relayed – ultimately transits the rain-faded 38 GHz backhaul to reach the core network (§III). c_{rain} leaves the other reward terms (fairness, outage, energy) untouched. Penalties (per user): severe outage $P_k^{\mathrm{sev}} = -100$ if $\mathrm{SINR}_{k,\mathrm{dB}} < -2$ dB; degraded service $P_k^{\mathrm{deg}} = -20$ if $-2 \leq \mathrm{SINR}_{k,\mathrm{dB}} < 0$ dB; and a shared low-battery penalty $P_{\mathrm{LB}} = -50$ for any platform with $\mathrm{SoC}_p < 10\,\%$ and at least one assigned user. The reported fairness metric (Jain index [28] over per-user delivered throughput, $T_k c_{\mathrm{rain}}$) differs from J_{fair} above, which is Jain over linear SINR and enters only the reward.

B. BATTERY MODEL

Each HAPS's battery state of charge (SoC) is tracked hourly,

$$\Delta\mathrm{SoC} = \frac{P_{\mathrm{solar}}(t) - (P_{\mathrm{backhaul}} + P_{\mathrm{payload}} + \sum_k P_{k,\mathrm{user}})}{C_{\mathrm{battery}}}, \tag{10}$$

with $P_{\mathrm{backhaul}} = 500$ W, $P_{\mathrm{payload}} = 200$ W, and $C_{\mathrm{battery}} = 12\,000$ Wh. Because the HAPS operates above >99 % of clouds, solar generation is independent of ground weather and follows a diurnal cosine profile,

$$I_0(t_{\mathrm{hr}}) = \begin{cases} 600\cos\big(\pi(t_{\mathrm{hr}} - 12)/12\big) & 6 \leq t_{\mathrm{hr}} \leq 18 \\ 0 & \text{night} \end{cases} \quad [\mathrm{W/m^2}], \tag{11}$$

giving $P_{\mathrm{solar}}(t) = 20\,\mathrm{m}^2 \times 0.20 \times I_0(t)$, i.e. 2400 W at noon and 0 W at night. SoC is clipped to $[0, 100]\,\%$; below 5 % the platform shuts down (no service), and below 10 % with assigned users the reward's low-battery penalty applies, so the agent must learn to balance load across platforms to avoid over-draining any one HAPS overnight.

IX. DQN RESULTS

The three stage-2 baseline policies (§VI-E: random, max-SINR, max-throughput) are now compared against the stage-3 LSTM+DQN agent (§VIII) on the same held-out days and six-action set. The DQN is a custom PyTorch (`torch/torch.nn`) network, trained on 400 rolling 24-hour episodes from 2024 and evaluated on 50 held-out 2025 days, across five experiments (random seeds).

LSTM+DQN has the highest reward and zero outage, with fairness statistically indistinguishable from max-throughput (Table 3). Max-SINR assumes perfect

TABLE 3: Policy comparison, 50 held-out 2025 days, mean over five experiments (std. dev. across experiments in parentheses). Jain is over per-user delivered throughput.

Policy	Reward/step	Jain	Outage/50
Random (six actions)	−40.5 (1.7)	0.563 (0.007)	16.5 (0.4)
Max-SINR (direct only)	−14.5 (2.1)	0.866 (0.002)	0.9 (0.2)
Max-throughput (six actions)	−3.8 (1.7)	**0.988** (0.000)	**0.0** (0.0)
LSTM + DQN	**+1.4** (1.0)	0.981 (0.006)	**0.0** (0.0)

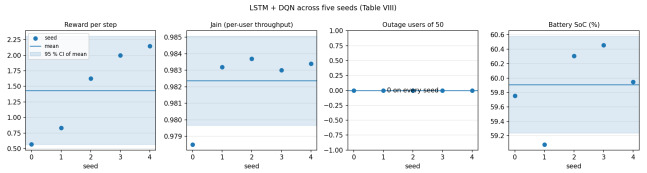


FIGURE 9: LSTM+DQN, five independent seeds (dot = one training run; line = mean; band = 95 % CI of the mean). Reward/step $+1.40 \pm 0.96$ (95 % CI $[0.21, 2.59]$); Jain 0.981 ± 0.006; outage 0.0 ± 0.0 on every seed and day; battery SoC 59.96 ± 0.59 %. The shaded band narrows with more seeds and reflects precision of the estimated mean, not run-to-run spread; zero outage and tight fairness/SoC variance across all five seeds show the policy is reliably learned, not a lucky single run.

SINR knowledge and cannot use the relay actions; max-throughput additionally assumes perfect, instantaneous knowledge of every option's SINR/capacity with no memory, battery or fairness awareness, so it is a theoretical ceiling rather than a deployable baseline – the DQN needs neither assumption. Max-throughput's ceiling is on the reward's throughput term T_k alone (Eq. (9)); it is not a ceiling on total reward, since its greedy, memoryless rule routes some users into the degraded-service and outage penalty bands that the DQN, trained on the full reward signal, learns to avoid, which is enough to flip the reward ranking despite max-throughput's higher raw throughput.

X. CONCLUSION

This paper characterised the HAPS–UAV–Gateway air-to-ground channel across its three link types, quantified multi-HAPS interference and the UAV-relay path as an alternative to the direct service link, and used that channel model to train and evaluate a DQN association agent.

Channel model. Spatial SINR heterogeneity – up to 9.94 dB at a fixed 50 km radius, driven by angular position relative to the two interfering HAPS – dominates service-link quality; the backhaul link remains noise-limited and largely unaffected by constellation geometry.

Relay path. HAPS→UAV→UE outperforms the direct link at every distance tested (11.1 vs. 2.7 bits/s/Hz at

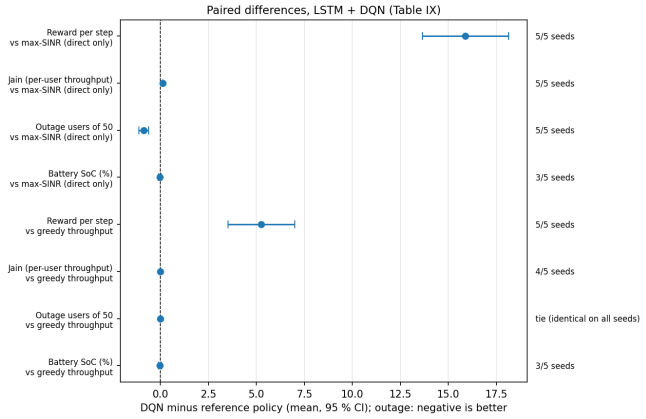


FIGURE 10: Paired differences, LSTM+DQN minus baseline, per experiment (50 days each); whiskers crossing zero are not significant. Vs. max-SINR: reward +15.87 ($p < 0.001$, 5/5), Jain +0.115 ($p < 0.0001$, 5/5), outage −0.87 ($p = 0.0007$, 5/5), battery SoC +0.03 ($p = 0.44$, tie). Vs. max-throughput: reward +5.17 ($p = 0.001$, 5/5), Jain −0.007 ($p = 0.07$, tie), outage 0 (tie), battery SoC ≈ 0 (tie). The DQN beats max-SINR on 3 of 4 metrics; against the stronger, information-privileged max-throughput baseline it wins only on reward, and ties – never loses – on fairness, outage and battery.

nadir); keeping the UAV near nadir on the 38 GHz feeder hop limits rain and scintillation losses, both of which grow sharply as the slant path lengthens at low elevation.

Learned association. The LSTM-fed DQN beats max-throughput and max-SINR on reward, with zero outage and no significant fairness loss relative to the information-privileged max-throughput upper bound.

Roadmap for future investigations. (i) Learn *when* relaying is worthwhile as a function of user distance; (ii) model movement of the HAPS and UAV platforms; (iii) consider more UAVs, including UAV selection – at fleet scale this should split into two separate MDPs (§VIII: UAV-to-HAPS assignment and UE-to-access-point assignment), with relay-vs-direct decided as its own explicit policy rather than folded into a single per-user choice, and with UAV placement itself chosen to avoid the blind-corner coverage gaps a naive non-overlapping tiling leaves behind, reducing the resulting edge-of-coverage errors; (iv) extend the LSTM forecast to neighbouring regions (e.g. a ConvLSTM on gridded reanalysis data, or a spatio-temporal graph model such as DCRNN, T-GCN or Graph WaveNet, if future sites are irregularly placed); (v) adapt to changing HAPS roles, shifting a node between serving and interfering based on its distance to the gateway, and jointly optimize UAV/HAPS positioning across both roles a node could play at once, rather than each in isolation; (vi) consider beamforming between HAPS and gateway; and (vii) explore forecast ensembling further – averaging the LSTM and persistence forecasts already improved RMSE in our initial check

(§VII).

XI. ABBREVIATIONS

ABS	Aerial Base Station
AGL	Above Ground Level
CDF	Cumulative Distribution Function
DRL	Deep Reinforcement Learning
DQN	Deep Q-Network
FDMA	Frequency Division Multiple Access
FSPL	Free-Space Path Loss
GW	Gateway
HAPS	High-Altitude Platform Station
ITU	International Telecommunication Union
K-band	18–27 GHz (ITU-R P.618 scintillation model)
Ka-band	27–40 GHz (Scaled from K-band via $(f/20)^{0.7}$)
LoS	Line-of-Sight
LSTM	Long Short-Term Memory
LWC	Liquid Water Content
MDP	Markov Decision Process
MLP	Multi-Layer Perceptron
MVP	Minimum Viable Product
NLoS	Non-Line-of-Sight
PDF	Probability Density Function
PL	Path Loss
RH	Relative Humidity
SINR	Signal-to-Interference-plus-Noise Ratio
SoC	State of Charge (Battery)
TDMA	Time Division Multiple Access
TD	Temporal Difference
UAV	Unmanned Aerial Vehicle
UTD	Uniform Theory of Diffraction

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